

Lecture(10)

Chapter(3)

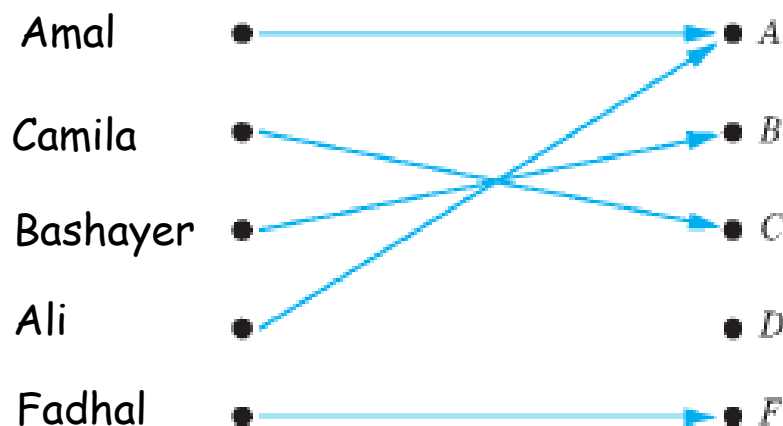
Function

- Introduction
- Definition of function
- Type of functions

Introduction

In many instances we assign to each element of a set a particular element of a second set (which may be the same as the first).

For example, suppose that each student in a discrete mathematics class is assigned a letter grade from the set $\{A, B, C, D, F\}$. And suppose that the grades are A for Amal, C for Camila, B for Bashayer, A for Ali, and F for Fadhal. This assignment of grades is illustrated as following



- This assignment is an example of a function and are just special kinds of relations.
- Function is extremely important in mathematics and computer science (for example, are used in the definition of such discrete structures as sequences and string, are used to represent how long it takes a computer to solve problems of a given size).

Definition of function

Definition:

Let A and B be nonempty sets. A *function* f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f : A \rightarrow B$.

Remark: Functions are sometimes also called **mappings** or **transformations**.

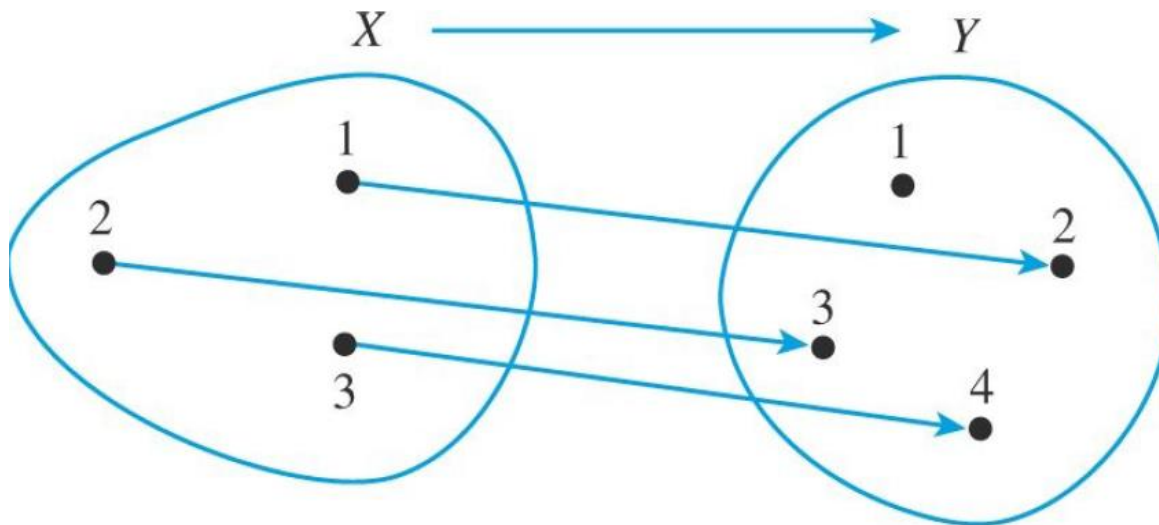
Definition: Suppose A and B are sets. A **function** f from A to B (denoted as $f : A \rightarrow B$) is a relation $f \subseteq A \times B$ from A to B , satisfying the property that for each $a \in A$ the relation f contains exactly one ordered pair of form (a, b) . The statement $(a, b) \in f$ is abbreviated $f(a) = b$.

$$\forall x \forall y_1 \forall y_2, \quad (((x; y_1) \in f \wedge (x; y_2) \in f) \Rightarrow y_1 = y_2)$$

Definition of function

Example Let $X = \{1, 2, 3\}$ and $Y = \{1, 2, 3, 4\}$. The formula $f(x) = x + 1$

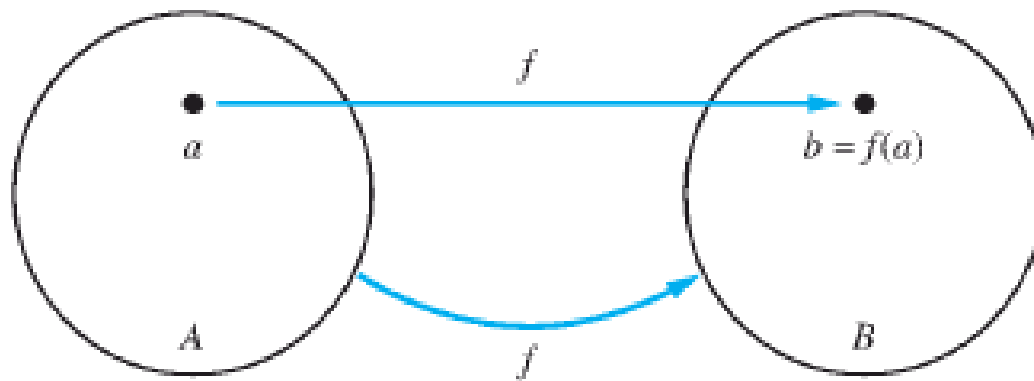
Defines as function $f: X \rightarrow Y$. For this function, $f(1) = 2$, $f(2) = 3$, and $f(3) = 4$.



Definition of function

Definition:

If f is a function from A to B , we say that A is the *domain* of f and B is the *codomain* of f . If $f(a) = b$, we say that b is the *image* of a and a is a *preimage* of b . The *range*, or *image*, of f is the set of all images of elements of A . Also, if f is a function from A to B , we say that f maps A to B .



Definition of function

Example:

Suppose that each student in a discrete mathematics class is assigned a letter grade from the set $\{A, B, C, D, F\}$. And suppose that the grades are A for Amal, C for Camila, B for Bashayer, A for Ali, and F for Fadhal. What are the domain, codomain, and range of the function that assigns grades to students?

Solution:

Let G be the function that assigns a grade to student in our discrete mathematics class. Note that $G(\text{Amal}) = A$, for instance.

The domain of G is the set $\{\text{Amal}, \text{Camila}, \text{Bashayer}, \text{Ali}, \text{Fadhal}\}$.

The codomain is the set $\{A, B, C, D, F\}$.

The range of G is the set $\{A, B, C, F\}$, because each grade except D is assigned to some student.

Definition of function

Example:

Let R be the relation with ordered pairs $(\text{Amal}, 22)$, $(\text{Camila}, 24)$, $(\text{Bashayer}, 21)$, $(\text{Ali}, 22)$, and $(\text{Fadhal}, 24)$. Here each pair consists of a graduate student and this student's age. Specify a function determined by this relation.

Solution:

Example:

Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ assign the square of an integer to this integer. Then, $f(x) = x^2$, where the domain of f is the set of all integers, the codomain of f is the set of all integers, and the range of f is the set of all integers that are perfect squares, namely, $\{0, 1, 4, 9, \dots\}$.

Definition of function

Definition: Two functions $f : A \rightarrow B$ and $g : A \rightarrow D$ are **equal** if

$$f(x) = g(x) \text{ for every } x \in A.$$

Observe that f and g can have different codomains and still be equal. Consider the functions $f : \mathbb{Z} \rightarrow \mathbb{N}$ and $g : \mathbb{Z} \rightarrow \mathbb{Z}$ defined as $f(x) = |x| + 2$ and $g(x) = |x| + 2$. Even though their codomains are different, the functions are equal because $f(x) = g(x)$ for every x in the domain.

Definition:

Let f_1 and f_2 be functions from A to $\mathbf{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined for all $x \in A$ by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x),$$

$$(f_1 f_2)(x) = f_1(x) f_2(x).$$

Definition of function

Example:

Let f_1 and f_2 be functions from $\mathbf{R}$ to $\mathbf{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution: from the definition of the sum and product of function, it follows that

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

and

$$(f_1 f_2)(x) = x^2(x - x^2) = x^3 - x^4.$$

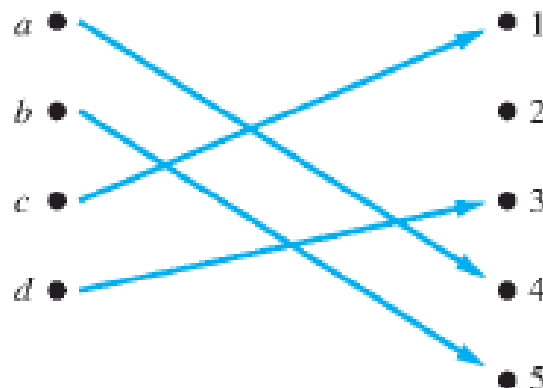
Type of functions

One-to-One function

Some functions never assign the same value to two different domain elements. These functions are said to be **one-to-one**.

Definition:

A function f is said to be *one-to-one*, or an *injection*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be *injective* if it is one-to-one.



Remark: We can express that f is one-to-one using quantifiers as $\forall a \forall b (f(a) = f(b) \rightarrow a = b)$ or equivalently $\forall a \forall b (a \neq b \rightarrow f(a) \neq f(b))$, where the universe of discourse is the domain of the function.

Type of functions

Example 11:

Suppose that each worker in a group of employees is assigned a job from a set of possible jobs, each to be done by a single worker. In this situation, the function f that assigns a job to each worker is one-to-one. To see this, note that if x and y are two different workers, then $f(x) \neq f(y)$ because the two workers x and y must be assigned different jobs. 

Example:

Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is one-to-one.

Solution:

Example:

Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

Solution:

Type of functions

Onto function

Definition:

A function f from A to B is called *onto*, or a *surjection*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called *surjective* if it is onto.

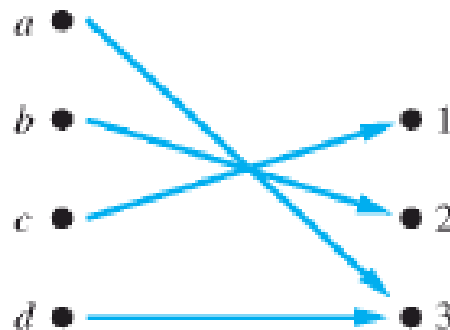
Remark: A function f is onto if $\forall y \exists x (f(x) = y)$, where the domain for x is the domain of the function and the domain for y is the codomain of the function.

Example:

Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function?

Solution:

Because all three elements of the codomain are images of elements in the domain, we see that f is onto.



Type of functions

Example:

Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

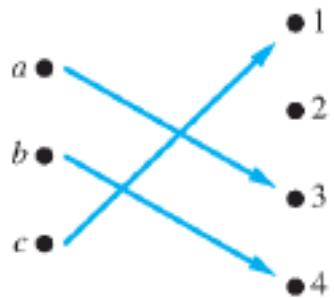
Solution:

Example:

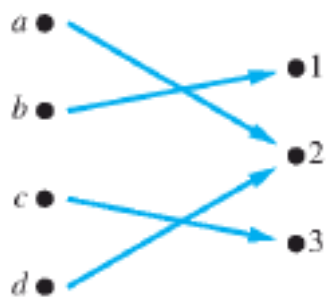
Consider the function f in Example 11 that assigns jobs to workers. The function f is onto if for every job there is a worker assigned this job. The function f is not onto when there is at least one job that has no worker assigned it. ◀

Examples of different types of correspondences

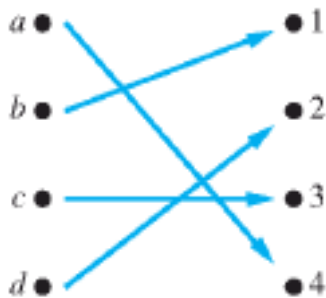
(a) One-to-one,
not onto



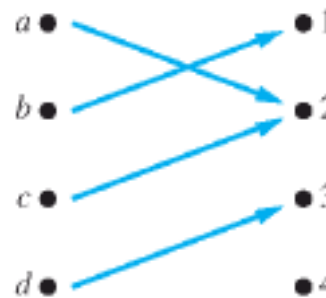
(b) Onto,
not one-to-one



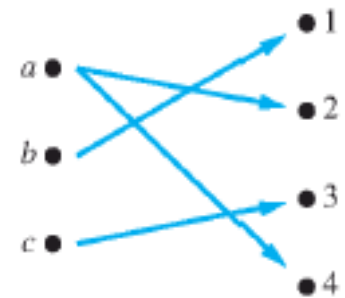
(c) One-to-one,
and onto



(d) Neither one-to-one
nor onto



(e) Not a function



Type of functions

Bijjective

Definition:

The function f is a *one-to-one correspondence*, or a *bijection*, if it is both one-to-one and onto. We also say that such a function is *bijjective*.

Example:

Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f a bijection?

Solution: The function f is one-to-one and onto. It is one-to-one because no two values in the domain are assigned the same function value. It is onto because all four elements of the codomain are images of elements in the domain. Hence, f is a bijection. 

Exercises

1. Why is f not a function from $\mathbf{R}$ to $\mathbf{R}$ if

a) $f(x) = 1/x$?

b) $f(x) = \sqrt{x}$?

c) $f(x) = \pm\sqrt{(x^2 + 1)}$?

2.

Suppose $A = \{0, 1, 2, 3, 4\}$, $B = \{2, 3, 4, 5\}$ and $f = \{(0, 3), (1, 3), (2, 4), (3, 2), (4, 2)\}$. State the domain and range of f . Find $f(2)$ and $f(1)$.

3.

Give an example of a relation from $\{a, b, c, d\}$ to $\{d, e\}$ that is not a function.

4.

A function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ is defined as $f(n) = 2n + 1$. Verify whether this function is injective and whether it is surjective.

5. Determine whether each of these functions from $\{a, b, c, d\}$ to itself is one-to-one and which functions are onto?

a) $f(a) = b, f(b) = a, f(c) = c, f(d) = d$

b) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$

c) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$

Exercises

6.

Determine whether the function $f(x) = x + 1$ from the set of real numbers to itself is one-to-one.

7.

Is the function $f(x) = x + 1$ from the set of integers to the set of integers onto?